

DOLITRAIL · DESIGN DECISIONS REGISTER

DoliTrail design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-04	Amish Chadha	O5 decided by Amish (round-3 decision 1A, DLT-DDR-004) and moved to Decisions made; no open decisions
0.2	2026-10-03	Amish Chadha	O1 to O4 decided by Amish (6A to 9A, DLT-DDR-003) and moved to Decisions made; new O5 on the wording of R7 and R8, proposed, awaiting Amish
0.1	2026-10-03	Amish Chadha	Register opened; design decisions made under Amish's 2026-10-03 pre-approvals; four requirement decisions proposed, awaiting Amish

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Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in `docs/REVIEW.md`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list decisions.

Safety: Several entries set safety limits (one wheel steadied with both hands, a parking lock on every kit, a solid axle, hand-tight clamps, sandbags before people, lift and carry on wet clay). Each takes the conservative option and names the evidence that would relax it. DoliTrail is not certified medical or rescue equipment.

■ Open decisions

None. O1 to O4 were decided on 2026-10-03 and O5 (the wording of R7 and R8) on 2026-10-04; all are under Decisions made.

■ To confirm when parts are bought

These are facts that can only be settled with real parts, real dolis or the first partner. None changes a decision; each may change a size or a limit.

Table 2. Items to confirm.

#	What to confirm	Why it matters	Source
1	Seller's ratings of the bought wheel (200 kg assumed) and tyre (150 kg); each wheel is proof-loaded to 285 kg before use (decision 6A)	R3	DLT-CAL-001, A10
2	Brake clamp force at about 150 N hand pull with the lock set (2.0 kN per pad assumed) and wet pad friction (0.35)	R5 brake factor	DLT-CAL-001, A5
3	Diametral crushing of local bamboo poles of 40 to 80 mm in the V jaws (4.0 kN for 40 mm length assumed)	R1 crushing factor, the hand-tight rule	DLT-CAL-001, A7
4	Rubber on wet bamboo friction (0.35 assumed)	R1 slip factor	DLT-CAL-001, A6
5	Tyre grip on the partner's trails, wet (0.35 assumed), and whether the 20 per cent carry limit on wet clay suits them	R5, decision 7A	DLT-CAL-001, A4, A13
6	How real dolis' beds are fixed to the poles; whether the lashing can be slid aside at the clamp points, and what to do with sewn sleeves	Clamp position; fitting time	DLT-DDR-002, A3
7	Pole spacing, diameter and length of dolis in the partner's villages	Arm range and width (R6)	DLT-DDR-001, D6
8	Masses of the bought wheel, brake, harness webbing and bag	R7	DLT-CAL-001, A12
9	That US7922183B2 (Hill-Rom stretcher brake) has lapsed, before any public claim about the brake	Design-around	DLT-PRC-001
10	Breaking strength of the 25 and 50 mm webbing bought for the hold-back strap and lift handles (5 and 10 kN assumed), and that the end caps stay on poles of 40 to 80 mm under a backward pull	Decisions 7A and 9A	DLT-CAL-001, A14

■ Value engineering

Value-engineering target: USD 1,500 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 482 (USD 1,018 under the target). Main cost drivers and savings worth trying:

- The largest lines are the two harnesses (USD 70), the fork unit (USD 62), the four side arms (USD 60), the wheel (USD 55) and the brake set (USD 45).
- Savings worth trying: cycle-rickshaw workshops already weld similar frames and could make the fork unit and arms together in a batch (perhaps USD 30 a kit); harnesses from used truck seat-belt webbing (about USD 25); a secondhand cargo wheel inspected and proof-loaded (about USD 25).
- Decisions 6A to 9A added USD 27 (hold-back strap USD 6, two lift handles USD 6, parts bag USD 15). If a bought wheel fails its proof load, the heavier tricycle wheel adds about USD 30.

■ Decisions made

Table 3. Decisions made.

Date	Decision	Decided by	Record
2026-10-03	TRL 2 review items D1 to D9: one wheel under the hips; rubber-lined V clamps on telescoping arms with hand bolts; cable disc brake with a parking lock on every kit; pole loops on slings; 20 inch cargo wheel with a solid axle; rated patient 120 kg on a 10 kg reference doli; first-trial kits kept in the ambulance; first co-design candidates (a district 108 ambulance operator in Mayurbhanj, a maternal health NGO in the same block, a volunteer mountain rescue group; none approached yet); R10 and R11 added, R1 to R9 targets unchanged; pitch, design-arounds and budget unchanged	Amish, pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." and "Proceed with the remaining 15 scaffolds"	DLT-DDR-001
2026-10-03	Design for construction, changes C1 to C11: welded fork unit with cross stubs and struts; telescoping side arms with V saddles; T-handle bolts and set screws; cross strap; caliper lobe on the dropout plate; lever mount; pole loops; solid axle in open slots; clamps on bare pole; wheel bag; pump and ties	Amish, same pre-approvals	DLT-DDR-002
2026-10-03	Assumptions A1 to A5: sound poles only; clamps hand tight, never with a bar (conservative; relaxed only by crush tests on local bamboo); sewn bed sleeves not fitted until the partner trial decides; bought ratings confirmed; jig-welded, inspected frame	Amish, same pre-approvals	DLT-DDR-002
2026-10-03	Sandbags before people: no person on the doli before the frame and wheel proof loads and the wet ramp test pass; the first person carried is a healthy volunteer (conservative; a gate, not relaxed)	Amish, same pre-approvals	DLT-BLD-001, sections 5 and 6
2026-10-03	O1, R3 (6A): keep the 20 inch cargo wheel; proof-load each bought wheel to 285 kg before use; heavier tricycle wheel and 20 x 2.4 cargo tyre only if it fails	Amish: "i agree with all the 46 recommendations you provided. please proceed."	DLT-DDR-003
2026-10-03	O2, R5 (7A): hold-back strap from the uphill bearer's hip belt to the rear pole ends; lift and carry on wet clay steeper than 20 per cent	Amish, same words	DLT-DDR-003
2026-10-03	O3, R7 (8A): the kit in two bags, about 7 and 8 kg, one per bearer	Amish, same words	DLT-DDR-003
2026-10-03	O4, R8 (9A): four people lift at steps, two at lift handles added to the frame, about 36 kg each	Amish, same words	DLT-DDR-003
2026-10-03	Appearance model departures: bamboo and cloth colours for the family's doli, a 1.75 m mannequin standing beside the doli on its far side, and a repeated front right clamp for the detail view, drawn for the renders only	Amish, same pre-approvals	docs/ REVIEW.md, TRL 3 section
2026-10-04		Amish: "For round 3, I agree with all your proposed recommendations"	DLT-DDR-004

Date	Decision	Decided by	Record
	O5, R7 and R8 (1A): R7 restated as "two bags of 10 kg or less, one per bearer"; R8 restated as "four people at steps, two at the lift handles"; both met as designed		